

Homework 1

Please show all work for full credit. You're only allowed to use the standard built-in Python 3 libraries.

Question 1. Watch [Mary's Room: A philosophical thought experiment – Eleanor Nelsen](#) on YouTube.

Do you think there is something about human learning that is different than machine learning?

Question 2. Rate your skills out of ten in the following subjects,

- 1) Probability and Statistics
- 2) Calculus
- 3) Linear Algebra
- 4) Python

Question 3. Of the sub-plots (a), (b), (c) and (d) in figure 1, state which ones have no correlation, negative correlation, positive correlation and non-linear correlation.

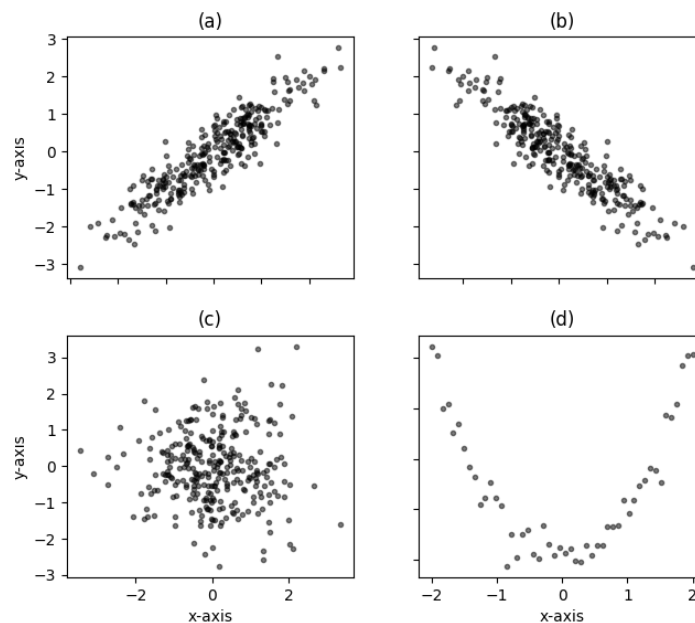


FIGURE 1—Different types of correlations.

Question 4. Assume that the probability of Alice going to class everyday given she got a good grade is $\frac{5}{6}$ and the probability of her getting a good grade is $\frac{2}{5}$ while the probability of her going to class everyday is $\frac{1}{3}$. What is the probability that Alice gets a good grade given she went to class everyday?

Hint: Thomas Bayes.

Question 5. We define the natural numbers as,

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$$

Let x be the arithmetic mean of the first $2^{33} - 1$ natural numbers. What is $\log_2(x)$?

Hint: Carl Friedrich Gauss.

Question 6. Let `arr = [x % 2 for x in range((2**100)-1)]` be a Python list using [Python list comprehension](#). What is the statistical mode of the list `arr`? Justify your answer.

Question 7. Define a function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$f(x) = \frac{\sin(x^2)}{y}$$

Calculate the following,

- The partial derivative with respect to y , $\frac{\partial f}{\partial y}$
- The partial derivative with respect to x , $\frac{\partial f}{\partial x}$
- The gradient vector $\nabla f(x, y)$.

Question 8. Let $\vec{v}_1 = [e, \pi, \sqrt{2}]$ and $\vec{v}_2 = [1, 2, 0]$. What is the dot product $\vec{v}_1 \cdot \vec{v}_2$? Please do not give an approximation.

Question 9. We define a matrix of n rows and p columns with real values as $\mathbf{A} \in \mathbb{R}^{n \times p}$. We can think of $\mathbf{A}$ as having n row vectors with the $i^{\text{th}} \leq n$ row vector being $\text{row}(\mathbf{A})_i$. Similarly, we can think of $\mathbf{A}$ as having p column vectors with the $j^{\text{th}} \leq p$ column vector being $\text{col}(\mathbf{X})_j$.

Then matrix multiplication between $\mathbf{A} \in \mathbb{R}^{n \times p}$ and $\mathbf{B} \in \mathbb{R}^{p \times m}$ is defined as,

$$\mathbf{AB} = \begin{bmatrix} \text{row}(\mathbf{A})_1 \text{col}(\mathbf{X})_1 & \text{row}(\mathbf{A})_1 \text{col}(\mathbf{X})_2 & \cdots & \text{row}(\mathbf{A})_1 \text{col}(\mathbf{X})_m \\ \text{row}(\mathbf{A})_2 \text{col}(\mathbf{X})_1 & \text{row}(\mathbf{A})_2 \text{col}(\mathbf{X})_2 & \cdots & \text{row}(\mathbf{A})_2 \text{col}(\mathbf{X})_m \\ \vdots & \vdots & \ddots & \vdots \\ \text{row}(\mathbf{A})_n \text{col}(\mathbf{X})_1 & \text{row}(\mathbf{A})_n \text{col}(\mathbf{X})_2 & \cdots & \text{row}(\mathbf{A})_n \text{col}(\mathbf{X})_m \end{bmatrix}$$

where $\text{row}(\mathbf{A})_i \text{col}(\mathbf{X})_j \in \mathbb{R}$ is a dot or inner product.

Further, the transpose of $\mathbf{A} \in \mathbb{R}^{n \times p}$ is written as $\mathbf{A}^T \in \mathbb{R}^{p \times n}$ and is defined as,

If $\mathbf{A}^T$ is the transpose of $\mathbf{A}$ then $\text{row}(\mathbf{A})_i = \text{col}(\mathbf{A}^T)_i$.

- What must be true of the number of columns of $\mathbf{A}$ and number of rows of $\mathbf{B}$ for $\mathbf{AB}$ to be defined?
- What information do the number of rows of $\mathbf{A}$ and number of columns of $\mathbf{B}$ give you about the dimensions of the product $\mathbf{AB}$?
- For the two matrices bellow, give the product $\mathbf{AB}$.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 4 & 2 & 2 \end{bmatrix}$$

- Prove that the matrix product is not a symmetric relation, i. e., $\mathbf{AB} \neq \mathbf{BA}$. Hint: can you construct a small counter example?
- Prove that $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$.
- For $\mathbf{X} \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$, $y \in \mathbb{R}^n$ prove that,

$$\beta^T \mathbf{X}^T y = y^T \mathbf{X} \beta$$

- Let $f(\beta) = \beta^T \beta$. Prove that the derivative $\frac{df}{d\beta} = 2\beta$.

Question 10. For $x \in \mathbb{R}$ we have,

$$f(x) = \frac{x^4}{4} - \frac{x^3}{3}$$

Give,

$$\min_x f(x) = \min_x \left(\frac{x^4}{4} - \frac{x^3}{3} \right)$$

Plot f to double check your answer.

Question 11. Let $x \sim \mathcal{N}(0, 1)$ be a normally distributed random variable with mean 0 and standard deviation 1. What is the likelihood that $x = \sqrt{2}$. Give an exact answer.

Question 12. The code snippet in listing 1 reads the plain text of the novel *The Cosmic Computer* by the famous American science fiction writer Henry Beam Piper over the internet and saves it to a string variable `text`.

```

1 from urllib.request import urlopen as get
2
3 url = 'https://www.gutenberg.org/files/20727/20727.txt'
4 with get(url) as response:
5     text = response.read().decode('utf-8')
6
7 print(text)

```

LISTING 1—A Python program to download the text of the *The Cosmic Computer* by Piper.

Write a Python program that prints out the ten most used words in the novel that have more than 5 letters. State your findings. Put your code in a file called `frequency.py`.

Question 13. Watch [Laziness in Python - Computerphile](#) on YouTube.

Implement the function `fibonacci(n)` in the code listing 2.

```

1 def fibonacci(n):
2     # implement me
3
4 for t in fibonacci(50):
5     print(t)

```

LISTING 2—A Python 3 program to print out the first $n \geq 0$ Fibonacci numbers.

Run your program for $n = 50$. What are the last five numbers printed? Put your code in a file called `fibonacci.py`.

SUBMISSION INSTRUCTIONS

- 1) Submit a PDF that answers all the questions.
- 2) Submit Python files, e. g., `frequency.py` and `fibonacci.py`.

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